

# Smart Metro Ticket Booking System

## Requirement Analysis Phase

|               |                            |
|---------------|----------------------------|
| Team ID       |                            |
| Project Name  | Smart Metro Booking System |
| Maximum Marks | 4 Marks                    |

### 3. Solution Requirements

This document consolidates the functional and non-functional requirements for the Metro Ticket Booking digitization solution built on ServiceNow, along with the component mapping, actor roles, assumptions, and delivery milestones agreed for this phase.

#### 3.1 Functional Requirements

##### Ticket Booking (Service Catalog)

- Provide a Service Catalog item for ticket booking with variables: Source Station, Destination Station, Passenger Type, and Number of Tickets.
- Validate station selection against the Station Master to prevent invalid routes.
- Support booking for multiple passenger types: General, Student, Senior Citizen.
- Allow booking of multiple tickets in a single transaction.

##### Fare Calculation

- Automatically calculate fare based on distance/zone between source and destination.
- Apply passenger-type discounts and multi-ticket totals automatically.
- Display an itemized fare breakup before payment confirmation.

##### Payment Processing

- Integrate digital payment modes: UPI, debit/credit card, and mobile wallets.
- Confirm payment status back to the booking flow in real time.
- Log every transaction with a unique reference number for audit and reconciliation.

##### QR Ticket Generation

- Generate a unique QR-coded digital ticket immediately after successful payment.
- Ensure each QR code is single-use and time-bound to the booked journey.
- Store the generated ticket against the passenger's booking record for later retrieval.

##### Notifications

- Send the digital ticket and QR code via in-app notification, email, and WhatsApp/SMS.
- Notify passengers of payment failures with a retry link.
- Alert Metro Operations of payment exceptions or QR generation failures.

##### Administration & Reporting

- Allow Station Managers to maintain station and fare master data through a controlled workflow.
- Provide Metro Operations with dashboards on ticket volume, revenue, and peak usage.
- Allow IT Admins to configure catalog variables, ACLs, and integration endpoints.

### End-to-End Solution Process Flow

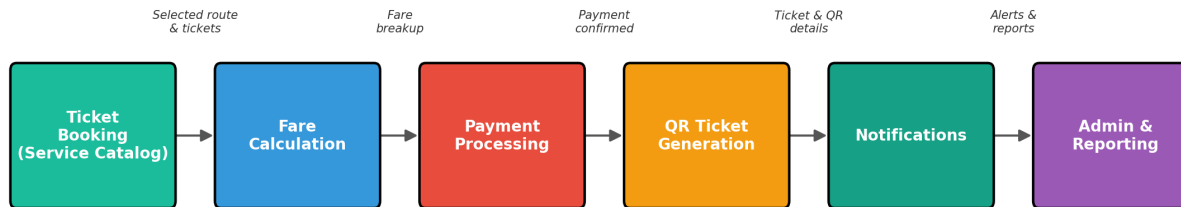


Figure 3.1 — End-to-end flow across the functional requirement areas above

## 3.2 Non-Functional Requirements

| Category        | Requirement                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|
| Performance     | Fare calculation and QR generation must complete within 3 seconds of payment confirmation under normal load.                    |
| Scalability     | The solution must support concurrent booking spikes during peak commute hours without degradation.                              |
| Availability    | Core booking and QR validation services should target 99.5% uptime during metro operating hours.                                |
| Security        | All payment data must be handled via PCI-compliant gateways; role-based access control (ACLs) must govern all sensitive tables. |
| Usability       | The booking flow should be completable by a first-time user in under 5 steps, with accessible, mobile-friendly UI Pages.        |
| Auditability    | Every booking, payment, and QR generation event must be logged with timestamp and correlation ID for traceability.              |
| Maintainability | Fare rules and station data must be configurable by IT Admins without requiring code deployment.                                |

## 3.3 ServiceNow Component Mapping

| Requirement Area    | ServiceNow Component                     | Purpose                                                                                        |
|---------------------|------------------------------------------|------------------------------------------------------------------------------------------------|
| Ticket Booking Form | Service Catalog + Catalog Client Scripts | Captures Source, Destination, Passenger Type, and Number of Tickets with real-time validation. |
| Fare Calculation    | Business Rules / Script Includes         | Executes fare logic server-side based on station distance and passenger type.                  |

| Requirement Area           | ServiceNow Component                          | Purpose                                                                                      |
|----------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------|
| Booking-to-Ticket Workflow | Flow Designer                                 | Orchestrates the end-to-end flow: payment trigger, QR generation, and notification dispatch. |
| Payment Integration        | Integration Hub / REST Spokes                 | Connects to external payment gateway APIs for UPI, card, and wallet transactions.            |
| QR Code Generation         | Integration Hub / Script Include (QR library) | Generates a unique QR image and links it to the ticket record.                               |
| Alerts & Ticket Delivery   | Notifications (Email/SMS/Push)                | Delivers digital tickets and status alerts to passengers and staff.                          |
| Passenger Self-Service     | Service Portal / UI Pages                     | Hosts the booking form, fare preview, booking history, and QR ticket display.                |
| Reporting                  | Performance Analytics / Reports               | Provides Metro Operations with usage, revenue, and peak-time dashboards.                     |

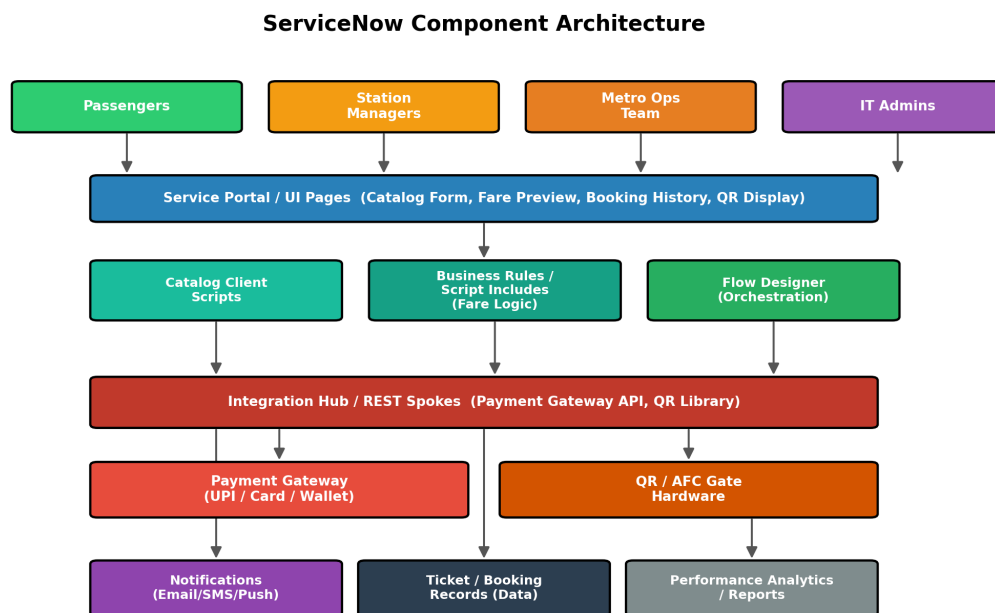


Figure 3.2 — Layered ServiceNow architecture realizing the component mapping above

### 3.4 Actors & Roles

| Actor                  | Role in the System                                                                                    |
|------------------------|-------------------------------------------------------------------------------------------------------|
| Passengers (End Users) | Book tickets, make payments, receive and use QR tickets, view booking history.                        |
| Station Managers       | Maintain station/fare master data, validate QR tickets at gates, monitor station-level ticket volume. |
| Metro Operations Team  | Consume analytics and reports, manage exceptions, plan schedules based on demand data.                |

| Actor     | Role in the System                                                                    |
|-----------|---------------------------------------------------------------------------------------|
| IT Admins | Configure catalog items, flows, integrations, ACLs, and troubleshoot platform issues. |

### 3.5 Delivery Milestones

| # | Milestone               | Description                                                                                               |
|---|-------------------------|-----------------------------------------------------------------------------------------------------------|
| 1 | Catalog Form Setup      | Build the Service Catalog item with Source, Destination, Passenger Type, and Number of Tickets variables. |
| 2 | Fare Logic Script       | Implement server-side fare calculation logic covering distance/zone and passenger-type rules.             |
| 3 | QR Code Integration     | Integrate QR generation library/API and link output to the ticket record.                                 |
| 4 | Station Details Mapping | Load and map the Station Master data used for route validation and fare lookup.                           |
| 5 | UI Pages                | Design and build passenger-facing Service Portal pages for booking, fare preview, and ticket display.     |
| 6 | UAT & Deployment        | Conduct user acceptance testing with all personas and deploy to production.                               |

### 3.6 Assumptions & Constraints

- Passengers have access to a smartphone or WhatsApp-enabled device to receive digital tickets.
- A payment gateway with UPI, card, and wallet support is available for integration.
- Station and fare master data will be provided/maintained by the Metro Operations Team.
- QR-based Automated Fare Collection (AFC) gate hardware is available or planned at stations for scanning.